

Formative Assessment 1.1

Topic 1 — SynapseOS: Designing a Conversational Interface Layer for Personal Computing

Despite decades of technological advancement, the dominant paradigm for personal computing remains largely unchanged — users still navigate desktops through icons, folders, and menus inherited from the 1980s. SynapseOS proposes a conversational interface layer that replaces the traditional GUI with a chat-based system powered by a language model, allowing users to interact with their operating system entirely through natural language. The vision is simple: if you can describe what you want, the system does it. No menus to learn, no commands to memorize, no interface to navigate — just intent. This research will design and evaluate this conversational layer built on top of an existing Linux base, assessing its viability, usability, and efficiency as a replacement for the conventional desktop experience across all types of users.

Topic 2 — UXL: A Syntax-Based Approach to Modeling UX Flows in Software Development

UX design and software development have long operated in parallel workflows with no shared formal language between them. Designers produce wireframes and user journey maps that developers must interpret subjectively, often leading to implementation gaps. This research proposes UXL, a domain-specific language with defined syntax for describing UX flows in a structured, human-readable, and machine-parseable format. By giving UX logic a formal grammar, UXL aims to bridge the communication gap between designers and developers, enable version-controlled UX documentation, and potentially allow automated validation of interaction flows during development.

Topic 3 — Crowdsourced Urban Traffic Mapping: A Participatory Sensing Platform for Real-Time Multi-Modal Awareness

Urban traffic management systems typically rely on fixed sensors and centralized data sources that are expensive, sparse, and slow to update. This research proposes a participatory sensing platform where everyday citizens contribute real-time traffic and mobility data through their devices, collectively building a live, city-wide awareness map that covers multiple transport modes including private vehicles, public transit, cyclists, and pedestrians. The study will investigate the design, feasibility, and data reliability of such a platform, exploring how crowdsourced input can supplement or replace traditional infrastructure in producing accurate, real-time urban mobility intelligence.